

Model inputs for ASPEN simulation

HeatX				
Name	CONDENS	EVAP1	EVAP2	TURBHEAT
Hot side property method	SRK	SRK	SRK	SRK
Hot side Henry's component list ID				
Hot side electrolyte chemistry ID				
Hot side use true species approach for electrolyt	YES	YES	YES	YES
Hot side free-water phase properties method	STEAMNB	STEAMNBS	STEAMNI	STEAMNBS
Hot side water solubility method	3	3	3	3
Cold side property method	SRK	SRK	SRK	SRK
Cold side Henry's component list ID				
Cold side electrolyte chemistry ID				
Cold side use true species approach for electrolyt	YES	YES	YES	YES
Cold side free-water phase properties method	STEAMNB	STEAMNBS	STEAMNI	STEAMNBS
Cold side water solubility method	3	3	3	3
Exchanger specification				
Units of exchanger specification				
Exchanger area [sqm]	150	50	70	20
Constant UA [J/sec-K]				
Minimum temperature approach [C]	6	5	5	5
Hot side outlet pressure [kPa]	0	0	0	0
Cold side outlet pressure [kPa]	0	0	0	0
EO Model components				
Hot side EO Model components				
Cold side EO Model components				
Inlet hot stream temperature [C]	68.4372	25	232.91	964.16159
Inlet hot stream pressure [kPa]	153.3253	101.325	101.325	101.325
Inlet hot stream vapor fraction	1	1	1	1
Outlet hot stream temperature [C]	31	24.052494	25	232.91045
Outlet hot stream pressure [kPa]	153.3253	101.325	101.325	101.325
Outlet hot stream vapor fraction	1	1	0.97672	1
Inlet cold stream temperature [C]	25	0.4670986	20	227.91045
Inlet cold stream pressure [kPa]	101.325	101.325	101.325	550
Inlet cold stream vapor fraction	0	1	1	1
Outlet cold stream temperature [C]	52.63654	20	26.7222	958.1183
Outlet cold stream pressure [kPa]	101.325	101.325	101.325	550
Outlet cold stream vapor fraction	0	1	1	1
Heat duty [Watt]	88662.03	46153.516	15889	52327.052
Calculated heat duty [Watt]	88662.03	46153.516	15889	52327.052
Required exchanger area [sqm]	150	50	70	20
Actual exchanger area [sqm]	150	50	70	20
Average U (Dirty) [Watt/sqm-K]	850	850	850	850
Average U (Clean)				
UA [J/sec-K]	8759.695	3852.1055	517.484	9505.0572
LMTD (Corrected) [C]	10.12159	11.981374	30.7044	5.5051802
LMTD correction factor	1	1	1	1
Thermal effectiveness				
Number of transfer units				
Number of shells in series	1	1	1	1
Number of shells in parallel				
Total feed stream CO2e flow [kg/min]	0	0	0.13203	1.4748084

Total product stream CO2e flow [kg/min]	0	0	0.13203	1.4748084
Net stream CO2e production [kg/min]	0	0	0	0
Utility CO2e production [kg/min]	0	0	0	0
Total CO2e production [kg/min]	0	0	0	0
Utility usage				
Utility cost				
Hot utility ID				
Cold utility ID				
Fan electricity utility ID				
Fan electricity usage				
Fan electricity cost				

Compr				
Name	FUEL	COM PUMP	COMF REGEN	TURB
Property method	SRK	SRK	SRK	SRK
Henry's component list ID				
Electrolyte chemistry ID				
Use true species approach for electrolytes	YES	YES	YES	YES
Free-water phase properties method	STEAMNB	STEAMNBS	STEAMNI	STEAMNBS
Water solubility method	3	3	3	3
Model Type	ISENTROP	ISENTROPI	ISENTRO	ISENTROPI
Specified discharge pressure [bar]	5.5		1	101.325
Specified pressure increase [bar]				
Specified pressure ratio				
Specified power required [Watt]				
Isentropic efficiency	0.9	0.9	0.9	0.9
Mechanical efficiency				
Polytropic efficiency				
EO Model components				
Indicated horsepower [Watt]	13084.56	98509.671	-71890.3	-39703.97
Calculated brake horsepower [Watt]	13084.56	98509.671	-71890.3	-39703.97
Net work required [Watt]	-26619.41	1.455E-11	-71890.3	-39703.97
Power loss [Watt]	0	0	0	0
Efficiency (polytropic / isentropic) used	0.9	0.9	0.9	0.9
Calculated discharge pressure [kPa]	550	153.32531	101.325	101.325
Calculated pressure change [kPa]	450	52.000311	52.0003	-448.675
Calculated pressure ratio	5.5	1.5132032	0.66085	0.1842273
Outlet temperature [C]	227.9104	68.437195	0.4671	964.16159
Isentropic outlet temperature [C]	208.0031	64.281251	-2.91415	906.14052
Vapor fraction	1	1	1	1
Displacement				
Volumetric efficiency				
Head developed [meter]	19392.4	3874.5736	-3490.84	-72647.62
Isentropic power requirement [Watt]	11776.1	88658.704	-79878.1	-44115.52
Inlet heat capacity ratio	1.39814	1.4009148	1.40161	1.2909563
Inlet volumetric flow rate [l/min]	3224.77	119388.62	80022.3	3432.4234
Outlet volumetric flow rate [l/min]	987.8416	89904.706	108905	13213.474
Inlet compressibility factor	0.99982	0.999861	0.99985	1.0009281
Outlet compressibility factor	1.002341	1.00021	0.99958	1.0002538
Compressor percent above surge				
Percent below stonewall				
Surge volume flow rate				

Stonewall volume flow rate				
Shaft speed				
Specific speed				
Inlet Mach number				
Total feed stream CO2e flow [kg/min]	1.342779	0	0	0.1320294
Total product stream CO2e flow [kg/min]	1.342779	0	0	0.1320294
Net stream CO2e production [kg/min]	0	0	0	0
Utility CO2e production [kg/min]	0	0	0	0
Total CO2e production [kg/min]	0	0	0	0
Utility usage				
Utility cost				
Utility ID				

RGibbs	
Name	REACTOR
Property method	SRK
Henry's component list ID	
Electrolyte chemistry ID	
Use true species approach for electrolytes	YES
Free-water phase properties method	STEAMNBS
Water solubility method	3
Specified pressure [bar]	0
Specified temperature [C]	4700
Specified heat duty [Watt]	0
EO Model components	
Outlet temperature [C]	1470.328
Outlet pressure [kPa]	550
Calculated heat duty [Watt]	0
Net heat duty [Watt]	0
Vapor fraction	1
Number of fluid phases	1
Maximum number of pure solids	0
Total feed stream CO2e flow [kg/min]	1.342779
Total product stream CO2e flow [kg/min]	0.132029
Net stream CO2e production [kg/min]	-1.21075
Utility CO2e production [kg/min]	0
Total CO2e production [kg/min]	-1.21075
Utility usage	
Utility cost	
Utility ID	

Mixer		
Name	MIX	WORKADD
Property method	SRK	SRK
Henry's component list ID		
Electrolyte chemistry ID		
Use true species approach for electrolytes	YES	YES
Free-water phase properties method	STEAMNB	STEAMNBS
Water solubility method	3	3
Specified pressure [kPa]	0	0
Temperature estimate [C]	25	
EO Model components		

Outlet temperature [C]	24.99839	
Calculated outlet pressure [kPa]	100	
Vapor fraction	1	
First liquid /Total liquid	1	
Total feed stream CO2e flow [kg/min]	1.342779	0
Total product stream CO2e flow [kg/min]	1.342779	0
Net stream CO2e production [kg/min]	0	0

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